



DESIGN GUIDELINES

SOCIAL ROBOT CO-DESIGN CANVASES

What things are important to consider in the robot's design?

Advantage guidelines

What advantages can the robotic solution have?

Think back to what you defined in the solution space canvas.

Prioritise advantages that reduce waiting times and administrative workload: automated self-check-in, guided data collection, and integration with vital-sign devices, while keeping interactions brief to improve Emergency Department patient flow.

Ethical guidelines

What ethical considerations does the robot have?

Think back to what you defined in the ethics canvas.

Protect patient privacy through encryption and minimum necessary data display; obtain informed consent by clearly explaining the AI-assisted nature of the system; ensure equitable access through multilingual and accessible design; and maintain human oversight by escalating high-risk or incomplete cases immediately to a nurse.

ROBOT DIMENSIONS

Environment guidelines

What should the robot's context be like?

For example:

- If users are especially vulnerable, should it optimize for support?
- If the robot is part of a strict process, should it optimize for efficiency and security?

The ED reception area is busy, noisy and used by highly vulnerable patients, so the kiosk should optimise for support and safety: durable, easy-to-clean, hospital-grade materials, positioned to preserve privacy while remaining visible and accessible.

Form guidelines

What guides the design of the robot's outward qualities?

For example:

- Should the robot be designed to appear especially approachable, or more industrial?
- Should it be simple, or detailed?

Favour an approachable, reassuring design over an industrial one: friendly visual cues, rounded edges, ergonomic and wheelchair-accessible construction, with clear visual and audio feedback throughout.

Interaction guidelines

What guides the design of interaction?

For example:

- Is the interaction multimodal, or is one modality optimized for efficiency?
- Should the user feel empowered and lead the interaction, or does the robot provide safety via leadership?
- Is the goal of the interaction to complete a task, or explore?

Support multiple modalities - touchscreen and voice - so patients can choose what suits them. Keep the interaction simple, brief and self-explanatory for first-time users; the goal is efficient completion of self-check-in, not exploration.

Behaviour guidelines

What guides the design of the robot's behaviour?

For example:

- Should behaviour be simple, or sensitive to context?
- Does the robot have internal drivers, or does it react to external stimuli?
- Does the robot have social skills?

Behaviour should be sensitive to context: detect when a patient needs help and offer an obvious way to request staff assistance. React to abnormal readings or technical faults by gracefully escalating to a nurse rather than continuing an unreliable assessment.

